

doing away with the need to measure everything.

Digital reconstructions of the brain tissue characterise a snapshot of the anatomy and physiology of the brain at a single moment in time. Blue Brain Project simulations use mathematical models of individual neurons and synapses to calculate the electrical activity of the network as it progresses over time. This demands a very high computational power that can only be delivered via large supercomputers. In fact, the larger the volume of tissue that needs to be simulated and higher the accuracy, the higher is the required computing power.

The range of experiments that can be enabled is proportional to the size and accuracy of digital reconstructions. Blue Brain Project is now constructing neuro-robotics tools wherein brain simulations are coupled to simulated robots and a simulated environment, in a closed loop. These novel tools enable replication of cognitive and behavioural experiments in animals, wherein the sensory organs capture and encode data about their environment, and their brain produces a motor response. The supercomputer based reconstructions and simulations put together by the project suggest a profoundly new strategy for comprehending the multilevel structure and function of the brain.

The Blue Gene Supercomputer

The Blue Brain workflow demands large-scale computing and data infrastructure. The hardware typically consists of the following configuration:

1. IBM 65,536-core BlueGene/Q supercomputer for modelling and simulation (hosted at CSCS),

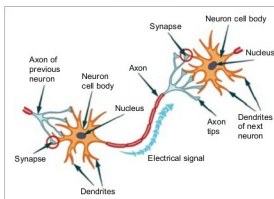


Fig. 3: Illustration showing the basic elements of the human brain (Image courtesy: www.study.com)



Fig. 4: BlueGene/L supercomputer used in the very early stages of the project (Image courtesy: www.ibm.com)



Fig. 5: BlueGene/Q is the latest supercomputer being used in Blue Brain Project (Image courtesy: www.ibm.com)

- providing extended capabilities for data-intensive supercomputing (Blue Gene Active Storage)

2. Fourty-node analysis and visualisation cluster

3. OpenStack/Ceph private Cloud running in two regions

4. Different storage systems for

data archiving and neuroinformatics

5. Modern continuous integration, collaborative software-development platform

State-of-the-art technology for acquisition of data on various brain levels of the brain organisation comprises multi-patch clamp set-ups for studies of the electrophysiological behaviour of neural circuits, multi-electrode arrays enabling the stimulation of and recording from brain slices, services for the formation and study of cell lines expressing specific ion channels, a multiplicity of imaging systems and systems for the 3D reconstruction of neural morphologies.

This infrastructure has been made available in partnership with EPFL's Laboratory for Neural Micro Circuitry. Accomplishment of Blue Brain Project hinges on very high volumes of standardised and high-quality experimental data that encompasses all likely levels of brain organisation. Data is sourced from the literature (via the project's automatic information extraction tools), from Human Brain Project, from large data-acquisition initiatives outside the project, and from EPFL's Laboratory for Neural Micro Circuitry.

Blue Brain workflow generates a huge need for computational power that falls in high-performance computing arena. In Blue Brain cellular-level models, depiction of detailed electrophysiology and communication of one neuron is estimated to require as high as 20,000 differential equations. While there are modern multi-core workstations, it is very challenging to solve such a high number of equations in biological real time.

Blue Brain Project's simulation of